

RoboPet - Living Room Background Wiring

Cursor task - paste this in and run it

Wire the new full-room pixel-art image in as the Living Room background, replacing the old wall sprite and the code-drawn SimpleFloor. Keep the dog on the lower floor and keep the pixels crisp.

1. Add the image as `assets/images/rooms/living_room_bg.png`. Confirm `assets/images/rooms/` is listed under `flutter: assets:` in `pubspec.yaml`.
2. In `LivingRoomWorld.onLoad`, remove the wall `SpriteComponent` that loads `rooms/living_wall.png` and remove the `SimpleFloor` instance.
3. Add the full-room background `SpriteComponent` from Section 2 (crisp-pixel paint, priority -10).
4. Set the dog start position to the lower floor and tune wander so it stays on the floor band (Section 4).
5. Run flutter pub get, hot restart, then verify every item in the Acceptance Test.

Acceptance test

- [] App opens to the Living Room and the new cyberpunk image fills the whole room (wall and floor in one image).
- [] The old flat SimpleFloor and any placeholder boxes are gone.
- [] The dog stands on the lower floor platform and wanders without walking up into the wall area.
- [] Background pixels are crisp (no blur) when scaled to the screen.
- [] The HUD (stat bars and currency) still renders on top and is unaffected.
- [] With `kDecorEnabled = false`, no Decorate button and no decor sprites appear.

1. Asset prep

- Save the image you generated as `assets/images/rooms/living_room_bg.png`.
- Open `pubspec.yaml` and confirm the rooms folder is registered:

```
flutter:  
  assets:  
    - assets/images/rooms/
```

- If it is already there from the earlier room work, no change is needed.

2. Replace the Living Room background

This is a single full-room image (wall plus floor baked in, no furniture), so it replaces both the old wall sprite and the SimpleFloor.

```
// At the top of the file, make sure Paint and FilterQuality are available.
import 'package:flutter/material.dart';
import 'package:flame/components.dart';

// Inside LivingRoomWorld.onLoad():
// 1) DELETE the old wall sprite line that loaded 'rooms/living_wall.png'.
// 2) DELETE the old SimpleFloor(...) line.
// 3) ADD the full-room background below.

// Room bounds (portrait, matches the source image ratio 768 x 1364).
final Vector2 kRoomTopLeft = Vector2(-200, -350);
final Vector2 kRoomSize = Vector2(400, 710);

final background = SpriteComponent(
  sprite: await Sprite.load('rooms/living_room_bg.png'),
  position: kRoomTopLeft,
  size: kRoomSize,
  anchor: Anchor.topLeft,
  priority: -10,
);
// Keep pixel art crisp when scaled up (nearest neighbor, no smoothing).
background.paint = Paint()
  ..filterQuality = FilterQuality.none
  ..isAntiAlias = false;
add(background);
```

3. Crisp pixels

The `filterQuality = FilterQuality.none` and `isAntiAlias = false` on the background paint stop Flutter from blurring the image when it scales the small pixel art up to screen size. Keep them on, otherwise the wall and floor will look soft and clash with the crisp dog sprite.

4. Dog placement and wander tuning

Put the dog on the lower floor platform, and keep its wander targets on the floor band so it never walks up into the wall.

```
// Place the dog on the lower floor platform.
add(RobotDog(Vector2(0, 250)));
```

If the dog drifts up onto the wall while wandering, clamp the chosen target inside `RobotDog` to the floor band:

```
// Walkable floor band in world coordinates.
const double kFloorMinY = 180; // top edge of the walkable floor
const double kFloorMaxY = 320; // bottom edge of the walkable floor

// When you pick a new wander target, clamp its Y:
target.y = target.y.clamp(kFloorMinY, kFloorMaxY);
```

Tune `kFloorMinY` / `kFloorMaxY` and the dog start Y until it sits naturally on the circular platform.

5. Notes

- Aspect ratio: source is about 768 x 1364 (ratio 1.776). `kRoomSize` 400 x 710 keeps it undistorted. If your camera frames the world differently, adjust `kRoomSize` and `kRoomTopLeft` together so the floor stays at the bottom.

- Decor: the room is empty, so a static background works now. You can re-enable interactive decor later by placing transparent decor sprites on top of this background.
- Other rooms: use the exact same pattern for Fabricator and Laboratory later, each with its own background image.
- PixelLab: save PixelLab credits for the dog sprite and animation frames (small, transparent), not for this large portrait background.